GAP-10D Θ-Hessian Principal-Symbol Decision: Fixed-Background Laplace Type, a Nonconic Resonance No-Go, and the Exact Composite Remainder

Ing. David Jaroš

1 August 2026

AI provenance — Tier B. AI-assisted; machine-verified against named sources or verifiers; individual attestation is not claimed. Details: `AI_PROVENANCE.md`.

Review profile. Machine verification: comprehensive for named claims; human review: selected principal claims; editorial approval: author approved. Registry: `PROVENANCE_REVIEW.yaml`.

Abstract

We perform the cheapest decisive test proposed for the induced-gravity route, but distinguish the quadratic Hessian of an action from the frozen self-consistency symbol of the D-composite equation. For the canonical Θ kinetic term with metric, connection and nondegenerate internal pairing held fixed, the raised quadratic Hessian has the scalar principal symbol $g^{\mu\nu}k_\mu k_\nu \mathbf{1}$. After Euclidean continuation it is a minimal Laplace-type operator, so the standard heat-kernel machinery applies to this fixed-background fluctuation problem. By contrast, the six-dimensional $q = 1$ sector of the frozen D-composite equation is a finite-scale singularity of the mixed-order full symbol $I - A(s, \lambda)$. Its singular surface is not conic under radial rescaling of the covector, and the first-order principal symbol $-A$ is generically rank nine. Therefore that six-dimensional resonance cannot be used directly as a Laplace bundle in a Gilkey calculation. The remaining first-principles task is now stated exactly: derive, gauge-fix and classify the full composite Θ-only Hessian including every chain-rule variation of $E[\Theta]$, $g[\Theta]$ and $\Omega[\Theta]$. No Einstein coefficient, physical mode count or unconditional GR derivation is claimed here. All symbolic statements are checked by `tools/verify_theta_hessian_principal_symbol.py`.

1 Scope and the two operators that must not be conflated

The working internal action contains a kinetic term of the schematic form

$$S_\Theta = \frac{1}{2}\int d^4x\sqrt{|g|}\,K_{AB}(D_\mu\Theta)^A(D^\mu\Theta)^B - \int d^4x\sqrt{|g|}\,V(\Theta), \tag{1}$$

where K_{AB} is a nondegenerate real pairing on the selected fluctuation space. There are two inequivalent linear objects in the present repository:

1. the Hessian $\mathcal{H} = \delta^2 S/\delta\Theta^2$ of an action;
2. the frozen self-consistency symbol $(I - A(s, \lambda))F = \partial\theta'$ of the torsion-free D-composite defining equation.

Only the first object is an input to a one-loop determinant. A singularity of the second object may identify a kinematic resonance, but it is not thereby a quadratic action or a Laplace-type Hessian.

2 Fixed-background Hessian

Hold $g_{\mu\nu}$, the background connection and the internal pairing fixed, and write $\Theta = \bar{\Theta} + \vartheta$. Let $D_\mu = \partial_\mu + C_\mu$ on the real fluctuation bundle. Up to boundary terms, the quadratic action has the form

$$S^{(2)}_{\text{fixed}} = \frac{1}{2}\int d^4x\sqrt{|g|}\left[K_{AB}(D_\mu\vartheta)^A(D^\mu\vartheta)^B + \vartheta^A U_{AB}\vartheta^B\right], \tag{2}$$

where U includes the potential Hessian and other derivative-free terms.

Theorem 1 (Fixed-background scalar principal symbol)**.** *If K is nondegenerate, the lower-index Hessian of Eq. (2) has*

$$\sigma_2(\mathcal{H}_{AB})(x,k) = K_{AB}\, g^{\mu\nu}(x)k_\mu k_\nu. \tag{3}$$

After raising one internal index with K^{-1},

$$\boxed{\sigma_2(\mathcal{H}^A{}_B)(x,k) = g^{\mu\nu}(x)k_\mu k_\nu\, \delta^A{}_B.} \tag{4}$$

The background connection contributes only first- and zero-order terms, and U contributes only zero order.

Proof. In the highest-derivative part of Eq. (2), replace D_μ by ∂_μ. Every insertion of C_μ lowers the number of derivatives acting on ϑ. Integration by parts therefore gives $-K_{AB}g^{\mu\nu}\partial_\mu\partial_\nu$ plus terms of order at most one. The potential Hessian is derivative-free. Replacing $\partial_\mu \mapsto ik_\mu$ and raising the internal index proves Eq. (4). The exact polynomial extraction is reproduced by the verifier. □

Corollary 2 (Conditional heat-kernel admissibility)**.** *After a controlled Euclidean continuation, the fixed-background operator can be written in minimal Laplace form*

$$\mathcal{H}_E = -\big(g_E^{\mu\nu}\nabla_\mu\nabla_\nu + \mathcal{E}\big). \tag{5}$$

Consequently the standard local heat-kernel coefficients apply to this specified fluctuation problem. This proves the operator-class assumption used in the companion induced-gravity calculation; it does not prove that the same operator is the full composite gravitational Hessian.

Proposition 3 (Metric-lock collapse of the same kinetic scalar)**.** *If the metric is no longer fixed but is defined by the same covariant jet,*

$$E_\mu = \mathcal{N}_0^{-1/2} D_\mu\Theta, \qquad \langle E_\mu, E_\nu\rangle_\sharp = g_{\mu\nu},$$

then in four dimensions the pointwise kinetic density obeys the exact identity

$$\frac{1}{2}\sqrt{|g|}\, g^{\mu\nu}\langle D_\mu\Theta, D_\nu\Theta\rangle_\sharp = 2\mathcal{N}_0\sqrt{|g|}. \tag{6}$$

Thus the fully metric-locked kinetic scalar is a volume term. Its composite second variation must be computed as the Hessian of the volume functional with the implicit D-composite constraints; it is not obtained by silently reusing the fixed-background Hessian of Theorem 1.

Proof. The metric lock gives $\langle D_\mu\Theta, D_\nu\Theta\rangle_\sharp = \mathcal{N}_0 g_{\mu\nu}$. Contracting with $g^{\mu\nu}$ gives $4\mathcal{N}_0$, and the prefactor one half yields Eq. (6). □

3 Why the six-dimensional resonance is not the principal bundle

The frozen D-composite analysis gives a 16×16 matrix $A(s, \lambda)$ linear in s, with

$$A^3 = qA^2, \qquad q = \lambda \cdot s, \qquad \det(I - A) = (1 - q)^6. \tag{7}$$

At $q = 1$, $\ker(I - A)$ is six-dimensional and has an injective curl map.

Theorem 4 (Nonconic resonance no-go)**.** *The singular set $q = 1$ of the full mixed-order symbol $I - A$ is not conic in covector space. In particular, if (s, λ) obeys $q = 1$, then under $s \mapsto cs$,*

$$\det\bigl(I - A(cs, \lambda)\bigr) = (1 - c)^6, \tag{8}$$

which is nonzero for generic $c \neq 1$. Therefore the $q = 1$ eigenspace cannot, as it stands, be the characteristic bundle of a homogeneous principal symbol. The actual first-order principal symbol of the differential system $I - A(\partial, \lambda)$ is $-A(s, \lambda)$; at generic exact points it has rank nine rather than full rank sixteen. It is neither a minimal scalar Laplace symbol nor an independently established six-component Laplace operator.

Proof. Linearity gives $A(cs, \lambda) = cA(s, \lambda)$. Substitution in the exact determinant identity yields the displayed radial dependence. Characteristic sets of homogeneous principal symbols are invariant under nonzero radial rescaling, whereas $q = 1$ is not. The exact generic rank-nine statement is verified directly from the same corrected $A(s, \lambda)$ used by the existing D-composite audit. □

Corollary 5 (Meaning of the $q = 1$ sector)**.** *The six modes are a finite-scale real-exponential resonance of the frozen self-consistency equation. They remain a potentially important clue to the connection/anholonomy sector, but they cannot be inserted directly as "six physical fields" in the induced-gravity trace. Their norm, statistics, gauge quotient and quadratic action must first be derived from the true Hessian.*

4 The exact remaining composite calculation

Let

$$\mathcal{F}[\Theta] = \bigl(E[\Theta], g[\Theta], \Omega[\Theta], \ldots\bigr) \tag{9}$$

denote every composite object substituted into a candidate single-Θ action. Schematically, the second variation of $S(\Theta, \mathcal{F}[\Theta])$ contains

$$\mathcal{H}_{\text{comp}} = S_{\Theta\Theta} + S_{\Theta\mathcal{F}}\mathcal{F}_{\Theta} + \mathcal{F}_{\Theta}^{*}S_{\mathcal{F}\Theta} + \mathcal{F}_{\Theta}^{*}S_{\mathcal{F}\mathcal{F}}\mathcal{F}_{\Theta} + S_{\mathcal{F}}\mathcal{F}_{\Theta\Theta}. \tag{10}$$

Equation (10) is schematic operator notation, but it records the terms that a fixed-background variation omits. Since E already contains a first derivative of Θ and a composite Levi–Civita connection contains derivatives of E, the differential order and degeneracy of $\mathcal{H}_{\text{comp}}$ cannot be inferred from Theorem 1.

The remaining decision procedure is now finite and explicit:

1. choose one candidate single-Θ action without an imported Hilbert–Palatini term;

2. compute Eq. (10) at a declared vacuum;

3. identify diffeomorphism, Lorentz and any internal zero modes;

4. add a declared gauge fixing and its ghost operator;

5. eliminate only algebraic auxiliary modes;

6. determine the differential order and the full principal symbol on the physical quotient;
7. if it is second-order and scalar Laplace type after Euclidean continuation, compute $\mathcal{E}$, the supertrace mode count and the a_2 coefficient.

A failure at step 6 is a genuine architecture result, not a failed heat-kernel calculation: it would force a nonminimal/pseudodifferential treatment or a revision such as the explicitly separated profile-field candidate.

Machine-verified B-tier status

GAP-10D-HESS-FIXED: CLOSED [L1]. The fixed-background quadratic Θ kinetic operator has the scalar principal symbol Eq. (4); its Euclideanized operator is of minimal Laplace type under the assumptions stated above.

GAP-10D-RES-PRINCIPAL: CLOSED AS NO-GO [L1]. The $q = 1$ six-dimensional D-composite resonance is not conic and cannot be identified directly with the principal-symbol bundle used in a Gilkey trace.

GAP-10D-HESS-COMP: NARROWED, not closed. The exact remaining lemma is the complete gauge-fixed second variation Eq. (10), followed by the physical symbol, mode and ghost count. This note does not derive the Einstein coefficient, Newton's constant, or an unconditional GR equation from the canonical single-Θ action.

These diagnostic subgap labels are B-tier machine-verified results. They do not alter the signed A-tier claims ledger until the author performs substantive review and explicitly promotes the wording.

References

[1] D. V. Vassilevich, "Heat kernel expansion: user's manual," *Phys. Rept.* **388**, 279–360 (2003).

[2] P. B. Gilkey, *Invariance Theory, the Heat Equation and the Atiyah–Singer Index Theorem*, 2nd ed., CRC Press (1995).